

Session: 4 query operators

Tasks

1.

Insert at least 5 documents into a MongoDB collection called playlists, where each document represents a Spotify playlist with fields: name (string), likes (number), genre (string), and followers (number).

Ans: insert 5 documents into playlists

```
db.playlists.insertMany([
```

```
{
```

```
  name: "Pop Hits 2025",
```

```
  likes: 2500,
```

```
  genre: "Pop",
```

```
  followers: 8000
```

```
},  
{  
  name: "Hip-Hop Vibes",  
  likes: 1800,  
  genre: "Hip-Hop",  
  followers: 6500
```

```
},  
{  
  name: "Chill Beats",  
  likes: 700,  
  genre: "Lo-Fi",  
  followers: 1200
```

```
},  
{  
  name: "Rock Legends",  
  likes: 950,
```

```
    genre: "Rock",
    followers: 3000
},
{
    name: "Workout Mix",
    likes: 450,
    genre: "EDM",
    followers: 900
}
])
```

Verify: `db.playlists.find().pretty()`

2.

Write a MongoDB query using the `$gt` operator to find all playlists with more than 1000 followers in the playlists

collection and display only the name and followers fields

Ans: find playlists with more than 1000 followers using \$gt

Display only name and followers fields:

```
Db.playlists.find(  
  {followers: { $gt: 1000}},  
  {_id:0, name:1, followwvers:1})
```

3.

Use the \$in operator to find all playlists where the genre is either 'Pop' or 'Hip-Hop' and sort the results by likes in descending order.

Ans: find playlists with genre 'pop; or 'hip-hop' using \$in

Sort by likes in descending order

```
Db.playlists.find({  
  Genre: { $in: ["pop", "hip-hop"]}  
}).sort({likes: -1})
```

4.

Query the playlists collection to find playlists with likes between 500 and 2000 (inclusive) using \$gte and \$lte, then limit the output to the top 3 playlists sorted by followers.

Ans: find playlists with likes between 500 and 2000

Use \$gte and \$lte, sort by followers and limit to top 3

```
Db.playlists.find({  
  Likes: {
```

```
$gte: 500,  
$lte: 2000}})  
.sort({ followers: -1})  
.limit(3)
```

5.

Suppose you want to recommend trending playlists like Spotify does. Write a MongoDB query that finds all playlists with more than 5000 followers and at least 1000 likes, but only show their name and genre. Explain in one line how you combined multiple query operators in your filter.

Ans: playlists with:

More than 5000 followers

At least 1000 likes

Show only name and genre:

```
Db.playlists.find({  
  Followers: {$gt: 5000},  
  Likes: {$gte: 1000}},  
  {  
    _id:0,  
    Name:1,  
    Genre:1})
```

> _MONGOSH

> db.playlists.find().pretty()

```
< {
  _id: ObjectId('6a2a568b9b0fc4ca3eb33ace'),
  name: 'Pop Hits 2025',
  likes: 2500,
  genre: 'Pop',
  followers: 8000
}
{
  _id: ObjectId('6a2a568b9b0fc4ca3eb33acf'),
  name: 'Hip-Hop Vibes',
  likes: 1800,
  genre: 'Hip-Hop',
  followers: 6500
}
{
  _id: ObjectId('6a2a568b9b0fc4ca3eb33ad0'),
  name: 'Chill Beats',
  likes: 700,
  genre: 'Lo-Fi',
  followers: 1200
}
{
  _id: ObjectId('6a2a568b9b0fc4ca3eb33ad1'),
  name: 'Rock Legends',
  likes: 950,
  genre: 'Rock',
  followers: 3000
}
```


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```
{
  _id: ObjectId('6a2a568b9b0fc4ca3eb33ad2'),
  name: 'Workout Mix',
  likes: 450,
  genre: 'EDM',
  followers: 900
}
> db.playlists.find(
  { followers: { $gt: 1000 } },
  { _id: 0, name: 1, followers: 1 }
)
< {
  name: 'Pop Hits 2025',
  followers: 8000
}
{
  name: 'Hip-Hop Vibes',
  followers: 6500
}
{
  name: 'Chill Beats',
  followers: 1200
}
{
  name: 'Rock Legends',
  followers: 3000
}
> db.playlists.find({
```

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```
    }  
> db.playlists.find({  
    genre: { $in: ["Pop", "Hip-Hop"] }  
}).sort({ likes: -1 })  
< {  
  _id: ObjectId('6a2a568b9b0fc4ca3eb33ace'),  
  name: 'Pop Hits 2025',  
  likes: 2500,  
  genre: 'Pop',  
  followers: 8000  
}  
{  
  _id: ObjectId('6a2a568b9b0fc4ca3eb33acf'),  
  name: 'Hip-Hop Vibes',  
  likes: 1800,  
  genre: 'Hip-Hop',  
  followers: 6500  
}
```

>_MONGOSH

```
}
> db.playlists.find({
  likes: {
    $gte: 500,
    $lte: 2000
  }
})
.sort({ followers: -1 })
.limit(3)
< {
  _id: ObjectId('6a2a568b9b0fc4ca3eb33acf'),
  name: 'Hip-Hop Vibes',
  likes: 1800,
  genre: 'Hip-Hop',
  followers: 6500
}
{
  _id: ObjectId('6a2a568b9b0fc4ca3eb33ad1'),
  name: 'Rock Legends',
  likes: 950,
  genre: 'Rock',
  followers: 3000
}
{
  _id: ObjectId('6a2a568b9b0fc4ca3eb33ad0'),
  name: 'Chill Beats',
  likes: 700,
  genre: 'Lo-Fi',
  followers: 1200
}
```

```
}  
> db.playlists.find(  
  {  
    followers: { $gt: 5000 },  
    likes: { $gte: 1000 }  
  },  
  {  
    _id: 0,  
    name: 1,  
    genre: 1  
  }  
)  
< {  
  name: 'Pop Hits 2025',  
  genre: 'Pop'  
}  
{  
  name: 'Hip-Hop Vibes',  
  genre: 'Hip-Hop'  
}  
insta_clone_db >
```